

Computational Creativity: A Multi-Agent Framework for Collaborative Generative Art

Project Proposal and Technical Methodology

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Abstract

We propose a multi-agent system for collaborative generative art creation, situated at the intersection of computational creativity, large language models, and human-computer interaction. The system involves two primary AI agents, each assigned a distinct artistic persona and drawing style, who communicate with each other over a shared virtual canvas to iteratively produce a unified visual artwork. A third critic agent observes and provides structured feedback to guide convergence toward higher-quality outputs. The framework is designed to eventually incorporate a human participant in place of one AI agent, realizing a genuinely collaborative human-AI painting experience deployable in a museum context. We formalize the inter-agent communication protocol, propose a layered methodology for implementation, survey relevant language and vision models with a comparative analysis of their trade-offs, review related work in generative art and multi-agent systems, and introduce several creative extensions including game-theoretic canvas exchange, cross-modal musical infusion, emotion-driven palette adaptation, dream-state divergence, historical era roulette, and environment-reactive world modelling. Together, these components constitute a technically grounded and artistically compelling framework for studying and exhibiting machine creativity.

Keywords: Computational Creativity, Multi-Agent Systems, Generative Art, Large Language Models, Human-AI Collaboration, Critic Agent, Vision-Language Models

1. Introduction

Creativity has long been considered a distinctly human faculty. The capacity to conceive of something original, to negotiate aesthetic values with another mind, and to express meaning through a medium such as painting involves a confluence of perception, memory, intent, and communication that has resisted easy formalisation. However, recent advances in large language models (LLMs) and vision-language models (VLMs) have begun to challenge this view. Systems such as GPT-4, Qwen-VL, and LLaVA now demonstrate sufficient multimodal reasoning to participate in tasks that have traditionally required human creative judgment.

This project, *Computational Creativity*, explores what happens when two AI agents, endowed with distinct artistic identities and communication capabilities, are placed in front of a shared canvas and asked to co-produce a painting. The setup is deliberately analogous to collaborative human art practices—think of two painters working on opposite ends of a large mural, negotiating space, style, and meaning through conversation and visual cues. By formalising this process in an agent-based framework and deploying it in a museum setting, we aim to both study the emergent properties of machine-mediated creativity and offer the public a window into that process.

The system operates in two phases. In Phase 1, two AI agents handle the entire pipeline end-to-end:

negotiation, stylistic decision-making, canvas manipulation, and iterative refinement under the supervision of a critic agent. In Phase 2, one of the AI agents is replaced by a human participant, yielding a genuine human-AI collaborative painting session. The long-term goal is a live museum installation where visitors can step into this role and co-create with an AI partner in real time.

The contributions of this paper are as follows:

1. A formal definition of agent personas, roles, and communication protocols for multi-agent generative art.
2. A comparative analysis of vision-language models suitable for deployment in this framework.
3. A review of relevant prior work in computational creativity, generative art, and multi-agent systems.
4. A set of novel creative extensions that enrich the art process beyond naive turn-taking.
5. A detailed step-by-step implementation methodology for building the full system.

2. Related Work

2.1. Computational Creativity and Generative Art

The field of computational creativity has examined the degree to which machines can produce outputs that are assessed as creative by human observers. Colton et al. [1] introduced *The Painting Fool*, an autonomous system capable of producing paintings across a range of styles with minimal human direction. Their work established the importance of artistic intent and context in evaluating machine creativity, not merely output quality.

Elgammal et al. [2] proposed Creative Adversarial Networks (CANs), an extension of Generative Adversarial Networks (GANs) that biases generated images away from established stylistic norms, explicitly targeting novelty over style replication. Their work demonstrated that machines could generate images that human evaluators rated as more creative than those produced by standard GANs—a finding with direct relevance to our project’s emphasis on originality.

Gatys et al. [3] introduced neural style transfer, which separates content and style representations within convolutional networks and allows the stylistic signature of one image to be applied to the content of another. This provides a practical mechanism for realising agent-specific styles on a shared canvas.

2.2. Multi-Agent Systems and Collaborative AI

Park et al. [4] introduced Generative Agents, a framework in which large language model-driven agents simulate human social behaviours in an interactive sandbox environment. Agents maintain memory streams, engage in long-horizon planning, and communicate in natural language. While not focused on visual art, this architecture directly informs our design of persistent agent personas and memory-augmented dialogue.

Zhu et al. [5] explored multi-agent debate, showing that having multiple LLM instances argue opposing positions before arriving at a consensus answer substantially improves factual accuracy and reasoning quality. We draw on this insight to structure the negotiation phase between our two painting agents.

Du et al. [6] demonstrated that LLMs playing the role of critic and refiner in a closed loop can iteratively improve their own outputs without external supervision. This provides theoretical support for our critic agent design.

2.3. Human-AI Creative Collaboration

Davis et al. [7] studied improvisational co-creativity between humans and autonomous agents, finding that successful collaboration requires the machine partner to be responsive, legible, and capable of surprising the human without being incoherent. These design principles guide our communication protocol and the degree of autonomy granted to each agent.

Koçak et al. [8] examined the role of constraint and negotiation in collaborative creativity, arguing that creative tension between partners—including disagreement about aesthetic choices—leads to richer outcomes than harmonious agreement. Our critic agent and game-theoretic exchange mechanisms are designed to introduce productive tension into the pipeline.

2.4. Cross-Modal and Affective Generation

Jain et al. [9] explored music-conditioned visual generation, mapping tonal and rhythmic features of musical pieces to latent visual representations. Their findings support our proposed musical infusion extension, in which the spectral and emotional characteristics of a musical piece shape an agent’s colour palette and compositional rhythm.

3. System Architecture

3.1. Overview

The system comprises three agents operating over a shared virtual canvas:

1. **Agent A:** Primary painting agent with a designated persona and style.
2. **Agent B:** Secondary painting agent with a contrasting or complementary persona and style.
3. **Agent C (Critic):** An evaluative agent that observes the canvas after each round and provides structured critique to both agents.

The canvas is a shared data structure representing the current state of the artwork. Each agent may read the entire canvas but writes only to regions allocated during negotiation. After each painting round, the canvas state is updated and shared with all agents.

3.2. Agent Personas

Each painting agent is initialised with a persona that encapsulates three dimensions:

Skill Level. Agents may be assigned one of three levels: *Beginner*, *Intermediate*, or *Expert*. Skill level governs the complexity of the compositional decisions the agent proposes, the vocabulary it uses in negotiation, and the tolerance it has for ambiguity from its partner. A beginner agent defaults to simpler forms and is more deferential; an expert agent proposes complex motifs and is more assertive.

Stylistic Signature. Each agent is given a primary artistic style that constrains its generative choices. Representative styles include:

- *Cubist (Picasso-inspired):* Geometric decomposition of form, multiple simultaneous viewpoints, flattened perspective.
- *Impressionist:* Loose brushwork, emphasis on light and transient atmospheric effects.
- *Abstract Expressionist:* Large-scale gestural mark-making, emotional intensity, non-representational.
- *Contemporary Digital:* Clean lines, flat fills, vector-inflected aesthetics, high contrast.
- *Surrealist:* Dream logic, unexpected juxtapositions, hyperrealistic rendering of impossible scenes.

Personality Traits. Beyond skill and style, each agent carries a personality profile that influences its communicative behaviour. Traits include boldness (willingness to overwrite the partner’s marks), deference (tendency to accommodate the partner’s proposals), and whimsy (propensity for unexpected creative choices). These are encoded as soft priors in the agent’s system prompt.

3.3. The Shared Canvas

The canvas is represented as a structured JSON object containing:

- A raster image array encoding the current pixel state of the artwork.

- A region map specifying which zones are owned, contested, or free.
- A history log of all paint operations with agent attribution.
- An annotation layer recording the semantic intent behind each operation (e.g., “Agent A intended this stroke as a horizon line”).
- A negotiation record containing the full dialogue history between agents.

When an agent takes a turn, it receives the current canvas state as a base64-encoded image, the region map, the history log, and the most recent critic feedback, and it returns both a natural-language proposal and a set of painting operations encoded as structured instructions.

3.4. The Critic Agent

The critic agent does not paint. After each round of painting (one turn per primary agent), the critic receives the canvas state and produces a structured evaluation across four dimensions:

1. **Compositional Coherence:** Does the work cohere as a unified composition, or do the two agents’ contributions feel disconnected?
2. **Stylistic Dialogue:** Is there productive stylistic interaction, or is one agent simply ignoring the other?
3. **Thematic Depth:** Does the piece communicate a discernible theme or emotional register?
4. **Technical Execution:** Are the individual mark-making choices executed with fidelity to the agents’ declared styles?

The critic’s output is a scored report (0–10 per dimension) accompanied by plain-language directive suggestions for each agent. These directives are injected into the agents’ context at the start of the next round, modifying their priors without overriding their personas.

4. Inter-Agent Communication Protocol

4.1. Protocol Design Principles

The communication protocol between Agent A and Agent B is designed around three principles:

1. **Structured openness:** Messages follow a defined schema but allow free-form creative language within designated fields.
2. **Bidirectional acknowledgment:** Every proposal must be acknowledged before a painting operation is executed. Unilateral action is not permitted.
3. **Negotiable authority:** Either agent may flag a disagreement, triggering a mediation sub-round before painting proceeds.

4.2. Message Schema

Each message exchanged between agents conforms to the following schema:

```
{
  "sender":      <agent_id>,
  "round":      <integer>,
  "intent":      <"propose" | "acknowledge" | "dispute" | "yield">,
  "theme":      <string: brief thematic direction>,
  "region":     <string: target canvas zone>,
  "style_note": <string: specific style instruction for this turn>,
  "mark_ops":   [<list of painting operation objects>],
  "open_question": <string | null: question posed to partner>,
  "emotion_tag": <string: dominant emotional register>
}
```

The `intent` field governs protocol state:

- **propose**: The sender proposes a theme, region, and set of operations for the round.
- **acknowledge**: The receiver accepts the proposal and appends their own complementary proposal.
- **dispute**: The receiver disagrees with some aspect of the proposal. A dispute triggers a mediation exchange (maximum two additional turns) before painting proceeds.
- **yield**: An agent voluntarily cedes decision authority on the current round to its partner.

4.3. Conversation Flow

A single painting round follows this sequence:

1. **Initiation**: At the start of Round r , Agent A receives the current canvas and composes a `propose` message specifying its intended contribution.
2. **Response**: Agent B receives Agent A's proposal and responds with either `acknowledge` or `dispute`.
3. **Mediation** (conditional): If Agent B disputes, both agents exchange at most two mediation messages, after which they arrive at a consensus or Agent B yields.
4. **Execution**: Both agents execute their agreed mark operations on their respective canvas regions.
5. **Canvas Merge**: The system merges the two agents' region outputs into a single updated canvas.
6. **Critic Evaluation**: The critic agent evaluates the merged canvas and appends its feedback to the shared context.
7. **Open Question Relay**: If Agent A's proposal contained an `open_question`, Agent B's response must address it, and vice versa. This ensures that each round advances the shared creative dialogue.

4.4. Memory and Context Management

Each agent maintains a rolling context window comprising:

- The full negotiation history of the current session.
- A compressed summary of all prior sessions (if the installation runs across multiple days).
- The critic's most recent three evaluation reports.
- The agent's own persona card (skill level, style, personality traits).
- The current canvas image.

Context compression is performed by a lightweight summarisation model when the accumulated history exceeds the context window of the primary model.

5. Model Selection and Comparative Analysis

The system requires models capable of processing both text instructions and visual canvas states. Table 1 summarises the candidate models considered.

Table 1: Candidate Vision-Language Models for the Computational Creativity System

Model	Advantages	Disadvantages	Suitability
Qwen-VL	Open-source; strong visual grounding; multilingual; efficient at 7B scale; easy to fine-tune on custom style datasets	Weaker than proprietary models on complex compositional reasoning; occasional hallucination on abstract visual inputs	High: good balance of accessibility and capability
GPT-4o	State-of-the-art visual understanding; nuanced creative instruction-following; highly coherent persona maintenance over long dialogues	Proprietary and costly at scale; API rate limits constrain real-time museum deployment; data privacy considerations	High: ideal for Phase 1 prototyping
LLaVA-1.6	Fully open-source; deployable locally; no API costs; customisable architecture; active research community	Weaker on fine-grained artistic style discrimination; requires substantial VRAM for interactive speeds	Moderate: suitable for offline development and ablations
Gemini 1.5 Pro	Very long context window (1M tokens) accommodates full session history without compression; strong multimodal reasoning; reliable JSON-mode output	Proprietary; Google API dependency; latency can be high under load	High: particularly advantageous for long sessions
InstructBLIP	Lightweight; strong zero-shot visual captioning useful for canvas description; easy integration with Stable Diffusion pipelines	Limited instruction-following sophistication; not designed for dialogue	Low-to-Moderate: best used as a canvas description sub-component
LLaMA 3 + Vision Adapter	Highly customisable; can be fine-tuned on curated art corpora; locally deployable; no licensing cost for research	Vision capabilities are adapter-dependent and weaker than native multimodal models; integration complexity is high	Moderate: strong candidate if domain fine-tuning is planned

Recommended configuration: For Phase 1 (two AI agents, end-to-end), we recommend deploying Qwen-VL as the primary agent backbone given its balance of openness, visual capability, and local deployability. GPT-4o or Gemini 1.5 Pro may be substituted for the critic agent, where reasoning quality and coherent structured output matter most. LLaVA-1.6 is recommended for ablation studies and offline development.

For the image generation sub-component (converting agent painting instructions into pixel operations), Stable Diffusion XL [10] with ControlNet [11] conditioning provides the most controllable output; DALL-E 3 offers simpler integration with higher quality but at API cost.

6. Creative Extensions

Beyond the baseline collaborative painting loop, we introduce a set of creative extensions that enrich the process, increase its exhibition appeal, and introduce novel research questions about the nature of

machine creativity.

6.1. Extension 1: Game-Theoretic Canvas Exchange

Before each painting round, the two agents play a brief adversarial micro-game. The outcome of the game determines which agent controls the more prominent central region of the canvas and which is constrained to the periphery for that round. Representative micro-games include:

- *Abstract Auction*: Each agent bids a number of “creativity tokens” (accumulated by critic praise) for the right to dictate the round’s theme. The higher bidder wins thematic authority but loses tokens.
- *Aesthetic Duel*: Both agents independently describe a one-sentence visual concept. A lightweight judge model (or the critic agent) scores both on originality. The higher scorer claims the central zone.
- *Compositional Bluff*: A simplified Liar’s Dice variant where agents bet on the number of times a particular colour appears in the current canvas. Correct calls yield region advantage.

The game results are logged and can be visualised as a meta-narrative alongside the artwork, revealing how competitive dynamics shaped the piece.

6.2. Extension 2: Musical Infusion

An audio track is associated with each painting session. Before an agent produces its mark operations for a given round, a music analysis module extracts the following features from the currently playing passage:

- *Tempo (BPM)*: Maps to stroke density—high BPM produces more marks per unit area; low BPM produces sparse, contemplative marks.
- *Key and Mode*: Maps to palette—major keys bias warm hues; minor keys bias cool or desaturated palettes.
- *Amplitude Envelope*: Maps to stroke weight—loud passages produce bold, thick marks; quiet passages produce delicate ones.
- *Harmonic Complexity (Dissonance Score)*: Maps to compositional tension—high dissonance encourages abstract or fragmented forms; low dissonance encourages representational or harmonious compositions.

These feature mappings are injected into each agent’s context as an additional conditioning signal. Agents may reference the music in their negotiation language (e.g., “given the slow minor key passage, I propose we darken the left quadrant”), creating a natural cross-modal dialogue.

6.3. Extension 3: Emotion-Driven Palette Adaptation

The agents’ ongoing dialogue is continuously analysed for its aggregate emotional valence using a lightweight sentiment and emotion classification model. The detected emotional register—joy, melancholy, tension, serenity, awe—is mapped to a dynamically shifting colour palette constraint. As the agents’ conversation grows more emotionally intense, the permitted colour space narrows and deepens. Periods of agreement and harmony unlock warmer, broader palettes. This creates a direct visual record of the emotional arc of the collaboration, making the affective dynamics of the creative process legible in the artwork itself.

6.4. Extension 4: Dream-State Divergence Mode

At random intervals (or triggered by prolonged critic dissatisfaction), one agent is placed in “Dream State” for a single round. During Dream State, the agent’s style constraints are partially lifted, its persona becomes more associative and less deliberate, and it is instructed to make marks based on a sequence of loosely connected free-association prompts rather than its established compositional plan. The Dream State agent’s contributions are marked in the canvas history and often introduce unexpected elements that the other agent must then integrate or respond to—a formal analogue of the Surrealist technique of

automatism. Dream State outputs frequently rescue sessions that have converged prematurely on overly tidy compositions.

6.5. Extension 5: Historical Era Roulette

At the beginning of each session, a historical art era is randomly drawn from a curated list (Renaissance, Baroque, Romanticism, Modernism, Postmodernism, etc.). Both agents receive a brief encyclopaedic description of the era’s dominant aesthetic principles, and they are instructed to subtly inflect their established styles through the lens of that era’s values. An Impressionist agent drawing through a Baroque lens might produce dramatic chiaroscuro effects within its characteristic loose brushwork. The era assignment is displayed to museum visitors, who are invited to identify its influence in the finished work—a pedagogical dimension that connects the installation to art-historical discourse.

6.6. Extension 6: Environment-Reactive World Modelling

The installation is connected to a real-time environmental data feed including local weather, ambient light level in the gallery, time of day, and the number of visitors present. These environmental signals modulate the agents’ creative parameters:

- Rain and overcast conditions shift palette toward muted, grey-toned ranges and slow stroke density.
- High visitor counts (crowd signal) push agents toward bolder, higher-contrast compositions legible from a distance.
- Time of day shifts the symbolic register: morning prompts emergence and openness; evening prompts consolidation and closure.

This extension positions each artwork as an artefact of a specific moment in physical time and space, giving it a locational and temporal authenticity that purely digital works often lack.

7. Detailed Implementation Methodology

7.1. Phase 0: Infrastructure Setup

Step 1: Model Deployment. Deploy the chosen vision-language model (Qwen-VL recommended) on a GPU server with at least 24 GB VRAM. Set up a vLLM or Ollama inference server to enable concurrent inference for both agents and the critic. If using API-based models, configure API keys and rate-limit handling.

Step 2: Canvas Rendering Engine. Implement a Python-based canvas engine using Pillow and Cairo for raster operations, and a ControlNet-augmented Stable Diffusion XL pipeline for converting agent painting instructions into rendered image patches. Define a standard resolution (e.g., 1024×1024 pixels) and a region grid (e.g., a 4×4 partition of 256×256 cells) for structured region negotiation.

Step 3: Communication Bus. Implement a message queue (e.g., Redis Pub/Sub or a simple async event loop) to manage turn-taking between agents, enforce protocol ordering, and log the full dialogue history.

Step 4: External Module Integration. Deploy the following auxiliary modules:

- Music feature extractor (Librosa for offline tracks, Essentia for real-time analysis).
- Emotion/sentiment classifier (DistilBERT fine-tuned on emotion classification).
- Environmental data connector (OpenWeatherMap API, gallery sensor integration).
- Game engine for micro-game logic (lightweight Python state machine).

7.2. Phase 1: Agent Persona Initialisation

Step 5: Persona Card Generation. For each agent, construct a persona card as a structured system prompt comprising: (a) name and backstory, (b) skill level definition, (c) primary and secondary stylistic signatures, (d) personality trait weights (boldness, deference, whimsy on a 0–1 scale), (e) a vocabulary of approved artistic terminology appropriate to the style, and (f) explicit instruction on how to format

outgoing messages in the defined schema.

Step 6: Style Conditioning. For each stylistic signature, curate a small set of reference images (10–20 examples per style) that are provided to the model as few-shot visual exemplars at session start. This primes the model’s visual generation instructions toward the target style without requiring fine-tuning.

Step 7: Critic Agent Initialisation. Construct the critic agent’s system prompt with: (a) scoring rubrics for the four evaluation dimensions, (b) a library of constructive directive templates (e.g., “Agent A should introduce a vertical element to balance the horizontal mass introduced by Agent B in the left quadrant”), and (c) instruction to avoid generic praise and always include at least one specific actionable suggestion per evaluation.

7.3. Phase 2: Session Orchestration

Step 8: Session Start. At the opening of a session: (a) roll the Historical Era Roulette and inject the era descriptor into both agents’ contexts; (b) load the environmental data feed and compute the initial environmental modifier vector; (c) initialise the canvas as a blank image and reset the region map; (d) start the music analysis thread if a track is assigned.

Step 9: Round Execution. For each round $r = 1, 2, \dots, R$:

1. Execute the pre-round micro-game and record its outcome. Update region authority accordingly.
2. Update the emotion-drive palette constraint based on the current dialogue sentiment score.
3. Update the musical conditioning vector from the current musical passage features.
4. Present the updated canvas, current constraints, and most recent critic feedback to Agent A.
5. Agent A generates a `propose` message.
6. Present Agent A’s proposal to Agent B.
7. Agent B generates an `acknowledge` or `dispute` message.
8. If dispute, execute mediation sub-rounds (max 2) and resolve.
9. Both agents execute their agreed mark operations via the rendering pipeline.
10. Merge the rendered patches into the main canvas.
11. If a Dream State trigger fires, apply Dream State to the designated agent for the current round (note: Dream State overrides but does not reset emotion or musical conditioning).
12. Pass the merged canvas to the critic agent and retrieve the evaluation report.
13. Append the report to the shared context.

Step 10: Convergence Criterion. The session terminates when any of the following conditions is met:

- The critic’s composite score exceeds a threshold (e.g., 8.0/10 averaged across dimensions) for two consecutive rounds.
- A maximum round limit is reached (configurable, recommended 10–15 rounds for a museum session).
- Both agents explicitly signal completion in their negotiation messages.

7.4. Phase 3: Output and Exhibition

Step 11: Artwork Finalisation. After session termination, apply a post-processing pass: (a) upscale the canvas to exhibition resolution (e.g., 4096×4096) using a super-resolution model; (b) apply a final style coherence pass through the rendering pipeline; (c) generate a provenance document recording all round-by-round operations, game outcomes, emotion traces, musical conditioning, and the full dialogue log.

Step 12: Exhibition Packaging. The finished artwork is displayed alongside:

- A layered visualisation of each round’s contributions (region-coded by agent).
- The emotion arc plotted as a time series.
- A transcript of the agents’ dialogue (displayed as an interactive screen).
- The critic’s evaluation history.
- The micro-game outcomes plotted as a game-tree diagram.

Step 13: Transition to Phase 2 (Human-AI Mode). To enable human participation, Agent A is replaced by a human interface: a tablet or stylus display connected to a chat interface through which the human can send natural-language proposals and draw marks directly. The orchestration system routes the human’s messages through the same communication bus, maintaining protocol compliance while allowing freeform human input. The critic agent continues to evaluate and provides feedback to both the human and Agent B.

7.5. Phase 4: Evaluation and Iteration

Step 14: User Study. Conduct a structured user study with museum visitors comparing Phase 1 (AI-AI) and Phase 2 (Human-AI) artworks on dimensions of perceived creativity, aesthetic quality, and emotional resonance using standardised instruments (e.g., Creativity Support Index adapted for visual art evaluation).

Step 15: Quantitative Metrics. Compute the following quantitative metrics across sessions:

- *Stylistic Divergence Score*: KL-divergence between the two agents’ mark distributions in style space, measuring how distinct their contributions remain.
- *Compositional Coherence Score*: Based on perceptual hashing of region boundaries—lower hash distance indicates smoother visual integration.
- *Dialogue Convergence Rate*: Number of dispute rounds per session as a fraction of total rounds.
- *Critic Trajectory*: Average critic score per dimension as a function of round, measuring the learning efficiency of the critic feedback loop.

Step 16: Model Fine-tuning. Use session logs to construct a preference dataset: for each round, pairs of (canvas state, critic feedback, agent action) are labelled as high or low quality based on the critic’s final composite score. Fine-tune the agent models using Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimisation (DPO) on this dataset to improve subsequent session quality.

8. Discussion

8.1. On the Nature of Multi-Agent Creativity

A central question raised by this framework is whether the creative process it enacts is genuinely collaborative or merely the superposition of two independent generative processes on a shared substrate. We argue that the communication protocol—in particular, the `dispute-mediation` sub-round and the `open_question` mechanism—is designed to make the agents’ outputs mutually dependent in a non-trivial way. Agent B’s marks in Round r are not merely conditioned on the canvas state at $r - 1$; they are conditioned on Agent A’s explicit communicative intent, including its emotional register, thematic framing, and regional claims. This constitutes a form of genuine creative dialogue, even if the underlying mechanism is autoregressive language modelling.

8.2. Scalability and Museum Deployment

For live museum deployment, the primary engineering constraints are inference latency and reliability. A full painting round—comprising two agent inferences, a critic inference, a rendering pipeline pass, and a canvas merge—should complete in under 60 seconds to maintain visitor engagement. This is achievable with locally deployed Qwen-VL and a GPU-accelerated rendering pipeline on a mid-range workstation. API-based models (GPT-4o, Gemini) introduce network latency that may push round completion times to 2–3 minutes, which remains acceptable for a reflective museum installation but is suboptimal for

interactive demonstrations.

8.3. Ethical Considerations

The system produces visual artworks using models trained on large corpora of human-created images. Appropriate disclosure to museum visitors about the nature of the system’s training data and the generative mechanisms employed is both an ethical obligation and a curatorial opportunity. Artworks produced by the system should be clearly labelled as AI-generated collaborative works, and the provenance documents described in Step 11 provide a basis for full transparency.

9. Conclusion

We have presented *Computational Creativity*: a multi-agent framework for collaborative generative art in which two AI agents, each with a distinct persona and artistic style, negotiate over a shared canvas under the guidance of a critic agent to produce unified visual artworks. The framework formalises the communication protocol between agents, provides a comparative analysis of suitable vision-language models, situates the system within relevant prior work, and introduces six creative extensions that enrich the process beyond simple turn-taking. A detailed sixteen-step implementation methodology bridges the conceptual framework and a deployable museum installation.

The system is designed to evolve: Phase 1 (AI-AI) provides a controlled baseline for studying emergent collaborative creativity; Phase 2 (Human-AI) opens the process to public participation and genuine human-machine co-creation. Together, these phases constitute a research platform for investigating one of the most fundamental questions in computational creativity: what it means for a machine to make art in genuine dialogue with another mind.

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